Chapter 10. Models for Matched Pairs

McNemar, symmetry, agreement, Bradley-Terry

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Matched pairs arise when each observation in one sample is paired with an observation in the other: the same subject measured twice, a case matched with a control, two raters judging the same slide. The two responses of a pair are dependent, and their cross-classification is a *square* $I \times I$ table with the same categories for rows and columns. This chapter develops the analysis of such tables: comparing the two margins (marginal models), modelling subject-specific effects (conditional models), loglinear models of symmetry type, agreement between raters, and paired preference (Bradley–Terry) models.

10.1 Comparing dependent proportions

Motivation. When the same people answer the same yes/no question twice, the two sample proportions are correlated, so the two-independent-sample formulas are wrong. We need the correct variance of a difference of dependent proportions, and the corresponding test of equal margins, which turns out to use only the discordant pairs.

For each of n matched pairs let π_{ab} be the probability of outcome a for the first observation and b for the second, n_{ab} the number of such pairs and $p_{ab} = n_{ab}/n$. We treat $\{n_{ab}\}$ as multinomial$(n; \{\pi_{ab}\})$. Then p_{a+} is the sample proportion in category a for observation 1 and p_{+a} for observation 2.

Definition (Marginal homogeneity)

The square table satisfies **marginal homogeneity** if $\pi_{a+} = \pi_{+a}$ for all a, i.e. the two observations have the same marginal distribution.

For a 2×2 table, $\pi_{1+} - \pi_{+1} = (\pi_{11} + \pi_{12}) - (\pi_{11} + \pi_{21}) = \pi_{12} - \pi_{21}$, so marginal homogeneity is equivalent to $\pi_{12} = \pi_{21}$: the table is symmetric about the main diagonal.

10.1.1 Inference for dependent proportions

Let $\delta = \pi_{+1} - \pi_{1+}$ and $d = p_{+1} - p_{1+} = p_{21} - p_{12}$.

Theorem 10.1 (Variance of a difference of dependent proportions)

Under multinomial sampling,

$$\begin{aligned} n \operatorname{Var}(d) &= \pi_{1+}(1 - \pi_{1+}) + \pi_{+1}(1 - \pi_{+1}) - 2(\pi_{11}\pi_{22} - \pi_{12}\pi_{21}) \\ &= (\pi_{12} + \pi_{21}) - (\pi_{12} - \pi_{21})^2. \end{aligned}$$

Hence $d \pm z_{\alpha/2}\,\hat{\sigma}(d)$ with $\hat{\sigma}^2(d) = \left[(p_{12} + p_{21}) - (p_{12} - p_{21})^2\right]/n$ is a large-sample confidence interval for δ.

Proof.

Step 1. Multinomial moments: $n \operatorname{Var}(p_{ab}) = \pi_{ab}(1 - \pi_{ab})$ and $n \operatorname{Cov}(p_{ab}, p_{cd}) = -\pi_{ab}\pi_{cd}$ for distinct cells.

Step 2. First form. $n \operatorname{Var}(p_{1+}) = \pi_{1+}(1 - \pi_{1+})$ and $n \operatorname{Var}(p_{+1}) = \pi_{+1}(1 - \pi_{+1})$ because each margin is binomial. For the covariance,

$$\begin{aligned} n \operatorname{Cov}(p_{+1}, p_{1+}) &= n \operatorname{Cov}(p_{11} + p_{21},\, p_{11} + p_{12}) \\ &= \pi_{11}(1 - \pi_{11}) - \pi_{11}\pi_{12} - \pi_{21}\pi_{11} - \pi_{21}\pi_{12} \\ &= \pi_{11}(1 - \pi_{11} - \pi_{12} - \pi_{21}) - \pi_{12}\pi_{21} = \pi_{11}\pi_{22} - \pi_{12}\pi_{21}. \end{aligned}$$

Then $\operatorname{Var}(d) = \operatorname{Var}(p_{+1}) + \operatorname{Var}(p_{1+}) - 2\operatorname{Cov}(p_{+1}, p_{1+})$ gives the first line.

Step 3. Second form. Directly, $d = p_{21} - p_{12}$, so

$$n\,\mathrm{Var}(d) = \pi_{21}(1-\pi_{21}) + \pi_{12}(1-\pi_{12}) + 2\pi_{12}\pi_{21} = (\pi_{12}+\pi_{21}) - (\pi_{12}-\pi_{21})^2.$$

Step 4. By the multivariate CLT $\sqrt{n}(d-\delta)$ is asymptotically normal, and $\hat{\sigma}(d)$ is consistent, so the interval follows from Slutsky's theorem. ■

The hypothesis of marginal homogeneity is $H_0 : \pi_{1+} = \pi_{+1}$ ($\delta = 0$). The ratio $z = d/\hat{\sigma}(d)$ is a Wald statistic. Under H_0 the variance simplifies to $(\pi_{12}+\pi_{21})/n$, estimated by

$$\hat{\sigma}_0^2(d) = \frac{p_{12}+p_{21}}{n} = \frac{n_{12}+n_{21}}{n^2}.$$

Theorem 10.2 (McNemar's test is the score test)

The score statistic for $H_0 : \pi_{12} = \pi_{21}$ in the multinomial model is z_0^2, where

$$z_0 = \frac{d}{\hat{\sigma}_0(d)} = \frac{n_{21}-n_{12}}{(n_{21}+n_{12})^{1/2}}; \tag{10.1}$$

z_0^2 is asymptotically χ_1^2 under H_0 (**McNemar's test**).

Proof.

Step 1. Reparametrize the four cell probabilities by $(\pi_{11}, \pi_{22}, \sigma, q)$ with $\sigma = \pi_{12}+\pi_{21}$, $q = \pi_{21}/\sigma$, so $\pi_{21} = \sigma q$, $\pi_{12} = \sigma(1-q)$, subject only to $\pi_{11}+\pi_{22}+\sigma = 1$. Marginal homogeneity is $q = \frac{1}{2}$.

Step 2. With $n^* = n_{12}+n_{21}$ the log-likelihood is

$$L = n_{11}\log\pi_{11} + n_{22}\log\pi_{22} + n^*\log\sigma + n_{21}\log q + n_{12}\log(1-q).$$

The parameter q does not enter the constraint and separates from $(\pi_{11}, \pi_{22}, \sigma)$, so the information matrix is block diagonal and the score test for q involves only the q-block.

Step 3. The score is $U(q) = n_{21}/q - n_{12}/(1-q)$ and, since $E(n_{21}) = n\sigma q$, $E(n_{12}) = n\sigma(1-q)$, the information is $\mathcal{I}(q) = n\sigma/[q(1-q)]$. At $q = \frac{1}{2}$ with σ replaced by its null ML estimate $\hat{\sigma} = n^*/n$:

$$U = 2(n_{21}-n_{12}), \qquad \mathcal{I} = 4n^*, \qquad \frac{U^2}{\mathcal{I}} = \frac{(n_{21}-n_{12})^2}{n_{21}+n_{12}} = z_0^2.$$

Step 4. Also $d/\hat{\sigma}_0(d) = [(n_{21}-n_{12})/n]/[(n_{12}+n_{21})^{1/2}/n]$, which is (10.1). The χ_1^2 limit is the usual score-test asymptotics. ■

Only discordant pairs matter for the test

McNemar's statistic depends only on the $n_{12}+n_{21}$ discordant pairs; the diagonal counts are irrelevant to *whether* the margins differ. All pairs do contribute, through n, to estimating *how much* they differ (d and its standard error).

10.1.2 Prime Minister approval rating example

Example 10.3 (Prime Minister approval)

A random sample of 1600 British voters rated the Prime Minister's performance twice, six months apart:

	Second survey		
First survey	Approve	Disapprove	Total
Approve	794	150	944
Disapprove	86	570	656
Total	880	720	1600

The sample odds ratio $794 \cdot 570/(150 \cdot 86) = 35.1$ shows strong association between the two surveys. The approval proportions are $p_{1+} = 0.59$ and $p_{+1} = 0.55$. By Theorem 10.1, $\hat{\sigma}(d) = 0.0095$ and a 95% interval for $\pi_{+1} - \pi_{1+}$ is $-0.04 \pm 1.96(0.0095)$, i.e. $(-0.06, -0.02)$. McNemar's statistic is

$$z_0 = \frac{86 - 150}{(86 + 150)^{1/2}} = -4.17 \qquad (z_0^2 = 17.4),$$

strong evidence that approval dropped.

10.1.3 Increased precision with dependent samples

For independent samples of size n each, $n\,\mathrm{Var}(\hat{\pi}_1 - \hat{\pi}_2) = \pi_1(1-\pi_1) + \pi_2(1-\pi_2)$, i.e. the first form in Theorem 10.1 without the covariance term. Matched data usually show *positive* dependence, $\log\theta = \log[\pi_{11}\pi_{22}/(\pi_{12}\pi_{21})] > 0$, and then the covariance term reduces $\mathrm{Var}(d)$. In Example 10.3, $\hat{\sigma}(d) = 0.0095$, while independent samples of size 1600 with the same proportions would give $\big[(0.59 \cdot 0.41 + 0.55 \cdot 0.45)/1600\big]^{1/2} = 0.0175$, almost twice as large. Within-subject designs thus estimate within-subject effects more precisely.

10.1.4 Small-sample test comparing matched proportions

Proposition 10.4 (Exact conditional test)

Conditional on $n^* = n_{12} + n_{21}$, $n_{21} \sim \mathrm{bin}(n^*, \pi_{21}/(\pi_{12} + \pi_{21}))$. Under $H_0 : \pi_{12} = \pi_{21}$ this is $\mathrm{bin}(n^*, \frac{1}{2})$, and the normal approximation to it gives exactly McNemar's z_0.

Proof.

Step 1. Collapse the multinomial to the three categories $\{(1,2), (2,1), \text{diagonal}\}$. Conditioning a multinomial on the total of a subset of cells gives a multinomial on that subset with renormalized probabilities; here the subset has two cells, so $n_{21} \mid n^*$ is binomial with success probability $\pi_{21}/(\pi_{12} + \pi_{21})$, which equals $\frac{1}{2}$ under H_0.

Step 2. The approximating normal has mean $n^*/2$ and variance $n^*/4$, so

$$z = \frac{n_{21} - \frac{1}{2}n^*}{(\frac{1}{4}n^*)^{1/2}} = \frac{n_{21} - n_{12}}{(n_{21} + n_{12})^{1/2}} = z_0.$$

■

For Example 10.3, $n^* = 236$ and the one-sided P-value $P(\mathrm{bin}(236, \frac{1}{2}) \geq 150) = 0.00002$; the two-sided P-value doubles this.

10.1.5 Connection between McNemar and Cochran–Mantel–Haenszel tests

Display each pair as its own 2×2 stratum: rows are the two observations (first, second), columns the two outcomes. Each stratum has row totals $(1, 1)$. For Example 10.3 there are 1600 strata of

four types (794 approve–approve, 570 disapprove–disapprove, 86 disapprove–approve, 150 approve–disapprove). We call this $2\times2\times n$ table the **subject-specific table**; the ordinary 2×2 table of pairs is the **population-averaged table**, since its margins estimate population marginal proportions. If, for each subject, the probability of success is the same at both observations, then occasion and response are conditionally independent given subject, which implies marginal homogeneity in the population-averaged table.

Theorem 10.5 (McNemar as a CMH statistic)

The Cochran–Mantel–Haenszel statistic for conditional independence in the $2\times2\times n$ subject-specific table equals $(n_{21}-n_{12})^2/(n_{21}+n_{12})$.

Proof.

Step 1. In stratum k let $c_k \in \{0,1,2\}$ be the number of successes (column-1 total). The CMH statistic is $\big[\sum_k (n_{11k} - En_{11k})\big]^2 / \sum_k \mathrm{Var}(n_{11k})$ with hypergeometric moments

$$E(n_{11k}) = \frac{1\cdot c_k}{2}, \qquad \mathrm{Var}(n_{11k}) = \frac{1\cdot 1\cdot c_k(2-c_k)}{2^2(2-1)} = \frac{c_k(2-c_k)}{4}.$$

Step 2. Concordant strata ($c_k = 0$ or 2) have $\mathrm{Var}(n_{11k}) = 0$ and $n_{11k} = E(n_{11k})$, so they contribute nothing.

Step 3. Discordant strata have $c_k = 1$, variance $\frac14$, and $n_{11k} - \frac12 = +\frac12$ for a (success, failure) pair (there are n_{12} of these) and $-\frac12$ for a (failure, success) pair (n_{21} of these). Hence

$$\mathrm{CMH} = \frac{\big[\frac12(n_{12}-n_{21})\big]^2}{\frac14(n_{12}+n_{21})} = \frac{(n_{21}-n_{12})^2}{n_{21}+n_{12}}.$$

■

The connection suggests how to treat more complex matched data: with several categories or several observations per subject, the generalized CMH tests with one stratum per subject test marginal homogeneity (Section 11.1.4 of Chapter 11).

10.2 Conditional logistic regression for binary matched pairs

Motivation. The tests of Section 10.1 correspond to models. Allowing each subject its own success probability gives a logit model with one nuisance parameter per subject; conditioning on their sufficient statistics removes them and yields the simple estimator n_{21}/n_{12} of a subject-specific odds ratio.

10.2.1 Marginal versus conditional models for matched pairs

Let (Y_1, Y_2) be the pair for a randomly selected subject, 1 = success, 0 = failure, and $x_1 = 0$, $x_2 = 1$.

Definition (Marginal and conditional models)

A **marginal** (population-averaged) model describes the marginal distributions, e.g.

$$P(Y_t = 1) = \alpha + \delta x_t, \qquad \mathrm{logit}\, P(Y_t = 1) = \alpha + \beta x_t.$$

A **conditional** (subject-specific) model lets the probabilities depend on the subject i:

$$\mathrm{link}\, P(Y_{it} = 1) = \alpha_i + \beta x_t, \qquad i = 1,\dots,n.$$

In the marginal logit model, the ML estimate is the log odds ratio of the marginal proportions, $\hat{\beta} = \log[p_{+1}p_{2+}/(p_{+2}p_{1+})]$. In the conditional model β describes the conditional association in the subject-stratified table.

Proposition 10.6 (Subject-specific versus population-averaged effects)

Let α_i be drawn from a distribution F in the conditional model.
(a) With the identity link, $P(Y_2 = 1) - P(Y_1 = 1) = \beta$; the subject-specific and population-averaged effects coincide.
(b) With the logit link, the population-averaged probabilities are $P(Y_t = 1) = E_F[e^{\alpha+\beta x_t}/(1+e^{\alpha+\beta x_t})]$, which in general is not of the form $e^{a+bx_t}/(1+e^{a+bx_t})$ with $b = \beta$.

Proof.

(a) Averaging $P(Y_{i2} = 1) - P(Y_{i1} = 1) = (\alpha_i + \beta) - \alpha_i = \beta$ over F gives β.

(b) The first statement is the law of total probability. For the second, take $\alpha = \pm c$ with probability $\frac{1}{2}$ each and $\beta > 0$. The marginal log odds ratio is $b = \operatorname{logit} E[\operatorname{expit}(\alpha + \beta)] - \operatorname{logit} E[\operatorname{expit}(\alpha)]$, and $E[\operatorname{expit}(\alpha)] = \frac{1}{2}$ by symmetry, so $b = \operatorname{logit}\left(\frac{1}{2}[\operatorname{expit}(c+\beta) + \operatorname{expit}(\beta - c)]\right)$. As $c \to \infty$ the bracket tends to $\frac{1}{2}(1+0)$, so $b \to 0$ while β is fixed. Hence $b \neq \beta$ in general. ■

10.2.2 A logit model with subject-specific probabilities

The model of Cox and Rasch is

$$\operatorname{logit} P(Y_{it} = 1) = \alpha_i + \beta x_t, \qquad x_1 = 0,\ x_2 = 1, \tag{10.2}$$

so for subject i, $P(Y_{i1} = 1) = e^{\alpha_i}/(1+e^{\alpha_i})$ and $P(Y_{i2} = 1) = e^{\alpha_i+\beta}/(1+e^{\alpha_i+\beta})$; for every subject the odds of success at observation 2 are e^{β} times those at observation 1. Given the parameters, responses are assumed independent, both across subjects and within a pair. Nevertheless the model induces dependence when averaged over subjects.

Proposition 10.7 (Heterogeneity induces positive association)

Suppose $\alpha_1, \ldots, \alpha_n$ are i.i.d. from F and, given α_i, Y_{i1}, Y_{i2} are independent with the probabilities of (10.2). Then $\operatorname{Cov}(Y_{i1}, Y_{i2}) \geq 0$ for every β, with equality if and only if F is degenerate.

Proof. [Supplement]

Step 1. By conditional independence, $E(Y_{i1}Y_{i2} \mid \alpha_i) = f(\alpha_i)g(\alpha_i)$ with $f(a) = \operatorname{expit}(a)$, $g(a) = \operatorname{expit}(a+\beta)$, so $\operatorname{Cov}(Y_{i1}, Y_{i2}) = \operatorname{Cov}\left(f(\alpha), g(\alpha)\right)$, $\alpha \sim F$.

Step 2. Let α' be an independent copy of α. Since f and g are both strictly increasing, $(f(\alpha) - f(\alpha'))(g(\alpha) - g(\alpha')) \geq 0$, and

$$0 \leq E[(f(\alpha) - f(\alpha'))(g(\alpha) - g(\alpha'))] = 2(E[fg] - E[f]E[g]) = 2\operatorname{Cov}(f(\alpha), g(\alpha)).$$

Step 3. Equality forces the nonnegative product to vanish a.s., i.e. $\alpha = \alpha'$ a.s. (strict monotonicity), which holds for independent copies only if F is degenerate. ■

The more variable the $\{\alpha_i\}$, the stronger the positive association: subjects with large α_i tend to succeed both times, subjects with small α_i to fail both times. So the model accounts for the dependence in matched pairs through its structure. The large number of $\{\alpha_i\}$, however, ruins ordinary ML (Proposition 10.9 below). The remedy is *conditional ML*, which conditions on sufficient statistics for the nuisance parameters. (Terminology: "conditional model" refers to subject-specific effects; "conditional ML" refers to the inference method.)

10.2.3 Conditional ML inference for binary matched pairs

Theorem 10.8 (Conditional ML estimator of the subject-specific odds ratio)

In model (10.2) let $S_i = y_{i1} + y_{i2}$.
(a) S_i is sufficient for α_i. Given $S_i = 0$ or $S_i = 2$ the pair is determined; given $S_i = 1$,

$$P(Y_{i1} = 0, Y_{i2} = 1 \mid S_i = 1) = \frac{e^\beta}{1 + e^\beta}, \qquad P(Y_{i1} = 1, Y_{i2} = 0 \mid S_i = 1) = \frac{1}{1 + e^\beta}.$$

(b) The conditional likelihood is $\ell_c(\beta) = (e^\beta)^{n_{21}}/(1 + e^\beta)^{n^*}$ with $n^* = n_{12} + n_{21}$, i.e. $n_{21} \mid n^* \sim \text{bin}(n^*, e^\beta/(1 + e^\beta))$.
(c) The conditional ML estimator and its standard error are

$$\hat\beta = \log \frac{n_{21}}{n_{12}}, \qquad \text{SE} = \Big(\frac{1}{n_{21}} + \frac{1}{n_{12}}\Big)^{1/2}.$$

Proof.

Step 1. Sufficiency. The joint mass function is

$$\prod_{i=1}^{n} \frac{e^{\alpha_i y_{i1}}}{1 + e^{\alpha_i}} \cdot \frac{e^{(\alpha_i+\beta) y_{i2}}}{1 + e^{\alpha_i+\beta}} \;\propto\; \exp\Big(\sum_i \alpha_i S_i + \beta \sum_i y_{i2}\Big),$$

an exponential family in which S_i is sufficient for α_i; so the conditional distribution given $\{S_i\}$ is free of $\{\alpha_i\}$.

Step 2. Conditional probabilities. Given $S_i = 0$ ($S_i = 2$) the only possible pair is $(0, 0)$ ($(1, 1)$), which carries no information about β. Given $S_i = 1$, write $c_i = (1 + e^{\alpha_i})(1 + e^{\alpha_i+\beta})$. Then

$$P(0, 1) = \frac{e^{\alpha_i+\beta}}{c_i}, \quad P(1, 0) = \frac{e^{\alpha_i}}{c_i}, \quad P(0, 1 \mid S_i = 1) = \frac{e^{\alpha_i+\beta}}{e^{\alpha_i} + e^{\alpha_i+\beta}} = \frac{e^\beta}{1 + e^\beta}.$$

Step 3. Conditional likelihood. Multiplying over the n^* discordant pairs, n_{21} of which are $(0, 1)$ and n_{12} are $(1, 0)$, gives $\ell_c(\beta) = e^{\beta n_{21}}/(1 + e^\beta)^{n^*}$; as a sum of n^* independent Bernoulli variables, n_{21} is binomial.

Step 4. Maximization. $\log \ell_c = n_{21}\beta - n^* \log(1 + e^\beta)$ has derivative $n_{21} - n^* e^\beta/(1 + e^\beta)$, which vanishes when $e^\beta/(1 + e^\beta) = n_{21}/n^*$, i.e. $e^{\hat\beta} = n_{21}/n_{12}$; the log-likelihood is strictly concave, so this is the maximum.

Step 5. Standard error. The observed information is $n^* e^\beta/(1 + e^\beta)^2$; at $\hat\beta$ it equals $n^*(n_{21}/n^*)(n_{12}/n^*) = n_{21}n_{12}/n^*$, whose inverse is $n^*/(n_{21}n_{12}) = 1/n_{21} + 1/n_{12}$. ∎

Testing $\beta = 0$ conditionally is the binomial test with parameter $\frac{1}{2}$ of Proposition 10.4. When model (10.2) is realistic, this justifies comparing the margins using only the discordant pairs.

Proposition 10.9 (Inconsistency of unconditional ML)

Treating $\alpha_1, \ldots, \alpha_n$ as fixed parameters, the unconditional ML estimator of β in (10.2) is $2\log(n_{21}/n_{12})$, which converges to 2β.

Proof. [Supplement]

Step 1. For a concordant pair the likelihood is maximized by $\alpha_i \to \pm\infty$, with maximal value 1 whatever β is; these pairs drop out of the profile likelihood.

Step 2. For a $(1, 0)$ pair the likelihood is $p_1(1 - p_2)$ with $p_1 = \text{expit}(\alpha_i)$, $p_2 = \text{expit}(\alpha_i + \beta)$. Its α_i-derivative of the log is $(1 - p_1) - p_2$, which is zero when $p_1 + p_2 = 1$, i.e. $\alpha_i = -\beta/2$. The profiled value is $\text{expit}(-\beta/2)^2$. Similarly a $(0, 1)$ pair profiles to $\text{expit}(\beta/2)^2$.

Step 3. The profile log-likelihood $2n_{21}\log \operatorname{expit}(\beta/2) + 2n_{12}\log \operatorname{expit}(-\beta/2)$ has the form of Step 4 of Theorem 10.8 in the parameter $\beta/2$, so $\hat\beta/2 = \log(n_{21}/n_{12})$.

Step 4. By the law of large numbers $n_{21}/n^* \to e^\beta/(1+e^\beta)$, so $\log(n_{21}/n_{12}) \to \beta$ and $\hat\beta \to 2\beta$. ■

10.2.4 Random effects in binary matched-pairs model

Alternatively, treat $\{\alpha_i\}$ as a random sample from $N(\mu, \sigma^2)$ and integrate them out. The resulting marginal likelihood has only the three parameters (β, μ, σ); this is a *generalized linear mixed model* (Chapter 12). When the sample log odds ratio is nonnegative, it also yields $\hat\beta = \log(n_{21}/n_{12})$ (Neuhaus et al. 1994).

Proposition 10.10 (Mantel–Haenszel estimator)

Applied to the n subject-specific 2×2 tables, the Mantel–Haenszel estimator of the common odds ratio comparing observation 2 with observation 1 equals n_{21}/n_{12}.

Proof. [Supplement] Order the rows of stratum k as (observation 2, observation 1) and the columns as (success, failure), so $n_{11k} = y_{k2}$, $n_{12k} = 1 - y_{k2}$, $n_{21k} = y_{k1}$, $n_{22k} = 1 - y_{k1}$, and $n_{++k} = 2$. Then

$$\hat\theta_{MH} = \frac{\sum_k n_{11k}n_{22k}/n_{++k}}{\sum_k n_{12k}n_{21k}/n_{++k}} = \frac{\sum_k y_{k2}(1-y_{k1})}{\sum_k (1-y_{k2})y_{k1}} = \frac{n_{21}}{n_{12}}.$$

■

Thus the MH estimate, the conditional ML estimate and (with nonnegative log odds ratio) the random-effects ML estimate all give $e^{\hat\beta} = n_{21}/n_{12}$.

	$Y_2 = 1$	$Y_2 = 0$
$Y_1 = 1$	n_{11}	n_{12}
$Y_1 = 0$	n_{21}	n_{22}

shaded diagonal: $S_i \in \{0, 2\}$, no information on β
orange cells: $S_i = 1$, $n_{21} \mid n^* \sim \text{bin}\big(n^*, e^\beta/(1+e^\beta)\big)$

Figure 10.1. Only the discordant cells carry conditional information about the subject-specific effect β.

10.2.5 Logistic regression for matched case–control studies

The two members of a pair need not be the same subject. In a matched case–control study each case $(Y = 1)$ is matched with a control $(Y = 0)$ on characteristics that could affect the response, and both are measured on an explanatory variable X. For subject t in pair i consider

$$\text{logit}\, P(Y_{it} = 1) = \alpha_i + \beta x_{it}. \tag{10.3}$$

Although the design fixes Y and observes X, e^β is an XY odds ratio, which is the same for both conditional distributions, so it is estimable from retrospective data.

Example 10.11 (Diabetes and myocardial infarction)

A study of acute myocardial infarction (MI) among Navajo Indians matched 144 MI cases with 144 controls on age and gender, and recorded previous diagnosis of diabetes ($x = 1$ yes, 0 no):

	MI cases		
MI controls	Diabetes	No diabetes	Total
Diabetes	9	16	25
No diabetes	37	82	119
Total	46	98	144

In 37 pairs the case had diabetes and the control did not; in 16 pairs the reverse. The conditional ML estimate is $e^{\hat\beta} = 37/16 = 2.3$, so $\hat\beta = 0.84$.

10.2.6 Conditional ML for matched pairs with multiple predictors

With p predictors, $x_{it} = (x_{1it}, \ldots, x_{pit})'$, the model is

$$\operatorname{logit} P(Y_{it} = 1) = \alpha_i + \beta_1 x_{1it} + \cdots + \beta_p x_{pit} = \alpha_i + x_{it}'\boldsymbol{\beta}. \tag{10.4}$$

Theorem 10.12 (Conditional likelihood as no-intercept logistic regression)

In model (10.4), for pairs with $S_i = 1$,

$$P(Y_{i1} = 0, Y_{i2} = 1 \mid S_i = 1) = \frac{e^{x_{i2}'\boldsymbol{\beta}}}{e^{x_{i1}'\boldsymbol{\beta}} + e^{x_{i2}'\boldsymbol{\beta}}} = \frac{e^{x_i^{*\prime}\boldsymbol{\beta}}}{1 + e^{x_i^{*\prime}\boldsymbol{\beta}}}, \qquad x_i^* = x_{i2} - x_{i1}.$$

Hence:
(a) the conditional ML estimate of $\boldsymbol{\beta}$ is obtained by fitting an ordinary logistic regression, with no intercept, to the discordant pairs only, with response $y_i^* = 1$ for $(0,1)$ pairs, $y_i^* = 0$ for $(1,0)$ pairs and predictor x_i^*;
(b) a predictor with $x_{ji1} = x_{ji2}$ for all i (a between-pair effect) cancels from the conditional likelihood, so its coefficient cannot be estimated by conditional ML.

Proof.

Step 1. Exactly as in Step 2 of Theorem 10.8, with $c_i = (1 + e^{\alpha_i + x_{i1}'\boldsymbol{\beta}})(1 + e^{\alpha_i + x_{i2}'\boldsymbol{\beta}})$,

$$P(0,1 \mid S_i = 1) = \frac{e^{\alpha_i + x_{i2}'\boldsymbol{\beta}}/c_i}{(e^{\alpha_i + x_{i1}'\boldsymbol{\beta}} + e^{\alpha_i + x_{i2}'\boldsymbol{\beta}})/c_i} = \frac{e^{x_{i2}'\boldsymbol{\beta}}}{e^{x_{i1}'\boldsymbol{\beta}} + e^{x_{i2}'\boldsymbol{\beta}}};$$

dividing numerator and denominator by $e^{x_{i1}'\boldsymbol{\beta}}$ gives the logistic form in x_i^*. $P(1,0 \mid S_i = 1)$ is the complement.

Step 2. The conditional likelihood is $\prod_{S_i=1} \operatorname{expit}(x_i^{*\prime}\boldsymbol{\beta})^{y_i^*} [1 - \operatorname{expit}(x_i^{*\prime}\boldsymbol{\beta})]^{1-y_i^*}$, which is the likelihood of a no-intercept logistic regression of y^* on x^*. This proves (a).

Step 3. If $x_{ji1} = x_{ji2}$ for all i, then the jth component of every x_i^* is 0, so β_j does not appear in the conditional likelihood. This proves (b). ■

For Example 10.11, take $t = 1$ the control and $t = 2$ the case, so $y_i^* = 1$ always, and $x_i^* = -1$ (16 pairs), 0 (91 pairs), $+1$ (37 pairs). Pairs with $x^* = 0$ contribute a constant, and the no-intercept fit maximizes $37 \log \operatorname{expit}(\beta) + 16 \log \operatorname{expit}(-\beta)$, giving $\hat\beta = \log(37/16) = 0.84$, as before.

10.2.7 Marginal models and conditional models: extensions

Both kinds of model extend to multinomial responses and to matched *sets*: in (10.4), i may index a subject with repeated measurements or a cluster (children in a family, fetuses in a litter). By Theorem 10.12(b), conditional ML estimates only *within-cluster* effects, such as those in case–control and

crossover studies; information about a between-cluster effect is carried by subject totals, which are fixed by the conditioning. The random-effects approach (Chapter 12) has no such restriction. The rest of this chapter emphasizes marginal models and loglinear models for square tables; Chapter 11 extends marginal models to matched sets with covariates.

10.3 Marginal models for square contingency tables

Motivation. With $I > 2$ categories, comparing the two margins of an $I \times I$ table needs models that act directly on the marginal distributions: cumulative logits for ordinal responses and baseline-category logits (or Wald/score tests) for nominal ones.

For a square table $\{n_{ab}\}$ of outcomes (Y_1, Y_2) with $\pi_{ab} = P(Y_1 = a, Y_2 = b)$, marginal homogeneity is $P(Y_1 = a) = P(Y_2 = a)$, $a = 1, \ldots, I$.

10.3.1 Marginal models for ordinal classifications

With cumulative logits,

$$\operatorname{logit} P(Y_t \le j) = \alpha_j + \beta x_t, \qquad t = 1, 2, \quad j = 1, \ldots, I-1, \tag{10.5}$$

with $x_1 = 0$, $x_2 = 1$. This is a proportional odds structure: the odds of $Y_2 \le j$ are e^{β} times the odds of $Y_1 \le j$, for every j. The margins are stochastically ordered, $\beta > 0$ meaning that Y_1 tends to be higher than Y_2; marginal homogeneity is $\beta = 0$.

ML fitting must respect the dependence of (Y_1, Y_2): the likelihood is multinomial in the joint $\{\pi_{ab}\}$, but the model constrains only the margins π_{a+}, π_{+b}, so one cannot simply substitute the model into $\sum\sum n_{ab} \log \pi_{ab}$ (specialized constrained-ML algorithms are needed). Model (10.5) describes $2(I-1)$ marginal probabilities with I parameters, so its goodness-of-fit test has df $= I - 2$.

10.3.2 Premarital and extramarital sex example

Example 10.13 (Premarital and extramarital sex)

In the 1989 General Social Survey, 475 subjects rated premarital sex and extramarital sex on the scale 1 = always wrong, 2 = almost always wrong, 3 = wrong only sometimes, 4 = not wrong at all:

	Extramarital sex				
Premarital sex	1	2	3	4	Total
1	144	2	0	0	146
2	33	4	2	0	39
3	84	14	6	1	105
4	126	29	25	5	185
Total	387	49	33	6	475

The cumulative marginal proportions are $(0.307, 0.389, 0.611)$ for premarital and $(0.815, 0.918, 0.987)$ for extramarital sex; with scores $1, 2, 3, 4$ the means are 2.69 and 1.28. Model (10.5) gives $\hat{\beta} = 2.51$ (SE $= 0.13$): responses are much more permissive about premarital sex. Marginal homogeneity fits terribly ($G^2 = 348.1$, df $= 3$); model (10.5) has $G^2 = 35.1$ (df $= 2$), much better but still poor. Section 10.4.7 finds better models.

10.3.3 Marginal models for nominal classifications

For nominal responses, the baseline-category logit model

$$\log \frac{P(Y_t = j)}{P(Y_t = I)} = \alpha_j + \beta_j x_t, \qquad t = 1, 2, \quad j = 1, \ldots, I-1, \tag{10.6}$$

has $2(I-1)$ parameters for $2(I-1)$ marginal probabilities, so it is saturated; marginal homogeneity is $\beta_1 = \cdots = \beta_{I-1} = 0$. Maximizing the multinomial likelihood under this constraint (iteratively) and comparing fitted with observed counts by G^2 or X^2 tests marginal homogeneity with $\mathrm{df} = I-1$.

A non-iterative alternative uses the asymptotic normality of marginal differences. Let $d_a = p_{+a} - p_{a+}$ and $\mathbf{d} = (d_1, \ldots, d_{I-1})'$ (d_I is redundant since $\sum_a d_a = 0$).

Proposition 10.14 (Bhapkar and Stuart statistics)

(a) The covariance matrix $\mathbf{V}$ of $\sqrt{n}\,\mathbf{d}$ has elements

$$v_{ab} = -(\pi_{ab} + \pi_{ba}) - (\pi_{+a} - \pi_{a+})(\pi_{+b} - \pi_{b+}), \qquad a \neq b,$$
$$v_{aa} = \pi_{+a} + \pi_{a+} - 2\pi_{aa} - (\pi_{+a} - \pi_{a+})^2.$$

(b) Under marginal homogeneity the Wald (Bhapkar) statistic $W = n\mathbf{d}'\hat{\mathbf{V}}^{-1}\mathbf{d}$ is asymptotically χ^2_{I-1}. Stuart's statistic $W_0 = n\mathbf{d}'\hat{\mathbf{V}}_0^{-1}\mathbf{d}$ uses the null version $\hat{v}_{ab0} = -(p_{ab} + p_{ba})$, $\hat{v}_{aa0} = p_{+a} + p_{a+} - 2p_{aa}$, and

$$W = \frac{W_0}{1 - W_0/n}.$$

(c) For $I = 2$, W_0 is McNemar's statistic z_0^2.

Proof. [Supplement]

Step 1. Write $\mathbf{d}$ as the mean of i.i.d. vectors $\mathbf{X}_k$, one per pair, with $X_{ka} = \mathbb{1}\{Y_{k2} = a\} - \mathbb{1}\{Y_{k1} = a\}$. For $a \neq b$, the products $\mathbb{1}\{Y_2 = a\}\mathbb{1}\{Y_2 = b\}$ and $\mathbb{1}\{Y_1 = a\}\mathbb{1}\{Y_1 = b\}$ vanish, so

$$E(X_a X_b) = -P(Y_1 = b, Y_2 = a) - P(Y_1 = a, Y_2 = b) = -(\pi_{ba} + \pi_{ab}).$$

For $a = b$, $E(X_a^2) = P(Y_2 = a) + P(Y_1 = a) - 2P(Y_1 = Y_2 = a) = \pi_{+a} + \pi_{a+} - 2\pi_{aa}$. Subtracting $E(X_a)E(X_b) = (\pi_{+a} - \pi_{a+})(\pi_{+b} - \pi_{b+})$ gives (a).

Step 2. By the CLT, $\sqrt{n}(\mathbf{d} - E\mathbf{d}) \xrightarrow{d} N(\mathbf{0}, \mathbf{V})$, $\hat{\mathbf{V}} \xrightarrow{p} \mathbf{V}$, and under H_0, $E\mathbf{d} = \mathbf{0}$; so $W \xrightarrow{d} \chi^2_{I-1}$ (a quadratic form in an asymptotically $N(\mathbf{0}, \mathbf{V})$ vector with the inverse of a consistent estimate of $\mathbf{V}$).

Step 3. Comparing the sample versions, $\hat{\mathbf{V}} = \hat{\mathbf{V}}_0 - \mathbf{d}\mathbf{d}'$. By the Sherman–Morrison formula, with $q = \mathbf{d}'\hat{\mathbf{V}}_0^{-1}\mathbf{d} = W_0/n$,

$$\hat{\mathbf{V}}^{-1} = \hat{\mathbf{V}}_0^{-1} + \frac{\hat{\mathbf{V}}_0^{-1}\mathbf{d}\mathbf{d}'\hat{\mathbf{V}}_0^{-1}}{1 - q}, \qquad \mathbf{d}'\hat{\mathbf{V}}^{-1}\mathbf{d} = q + \frac{q^2}{1-q} = \frac{q}{1-q}.$$

Multiplying by n gives $W = W_0/(1 - W_0/n)$.

Step 4. For $I = 2$, $d_1 = p_{21} - p_{12}$ and $\hat{v}_{110} = p_{+1} + p_{1+} - 2p_{11} = p_{12} + p_{21}$, so $W_0 = n(p_{21} - p_{12})^2/(p_{12} + p_{21}) = (n_{21} - n_{12})^2/(n_{12} + n_{21})$. ∎

These tests use all $I-1$ degrees of freedom. With ordered categories, large I and strong dependence, a single-df ordinal test is often much more powerful.

10.3.4 Migration example

Example 10.15 (Migration 1980–1985)

Region of residence in 1980 and 1985 for a sample of 55,981 U.S. residents (NE = Northeast, MW = Midwest, S = South, W = West):

	Residence in 1985				
Residence in 1980	NE	MW	S	W	Total
NE	11,607	100	366	124	12,197
MW	87	13,677	515	302	14,581
S	172	225	17,819	270	18,486
W	63	176	286	10,192	10,717
Total	11,929	14,178	18,986	10,888	55,981

(The South–Midwest count is 225, the value consistent with the printed margins and fitted values.) About 95% of observations lie on the diagonal. The ML fit of marginal homogeneity has $G^2 = 240.8$ (df = 3), and Bhapkar's $W = 236.5$. The marginal proportions were $(0.218, 0.260, 0.330, 0.191)$ in 1980 and $(0.213, 0.253, 0.339, 0.194)$ in 1985. Applying Theorem 10.1 to the 2×2 table that collapses the other regions, a 95% interval for $\pi_{+1} - \pi_{1+}$ is $0.2131 - 0.2179 \pm 1.96(0.00054)$, i.e. -0.005 ± 0.001; for the other regions the intervals are -0.007 ± 0.001, 0.009 ± 0.001, 0.003 ± 0.001. The changes are statistically clear only because n is huge; they are small.

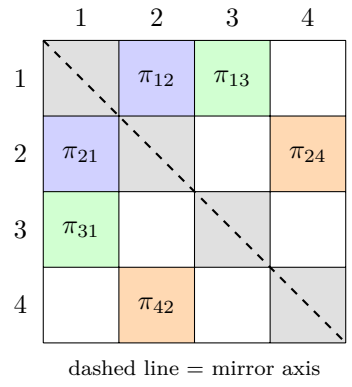
10.4 Symmetry, quasi-symmetry, and quasi-independence

Motivation. Instead of the margins one can model the joint distribution of the square table by loglinear models. Symmetry is too strict, but quasi-symmetry fits well, has a subject-specific interpretation, and yields a likelihood-ratio test of marginal homogeneity; quasi-independence describes the off-diagonal ("movers") structure.

Definition (Symmetry)

An $I \times I$ distribution satisfies **symmetry** if $\pi_{ab} = \pi_{ba}$ for all $a \neq b$.

Symmetry implies marginal homogeneity, since $\pi_{a+} = \sum_b \pi_{ab} = \sum_b \pi_{ba} = \pi_{+a}$. For $I = 2$ the two are equivalent; for $I > 2$ marginal homogeneity can hold without symmetry.



grey: diagonal, fitted exactly by S, QS, QI

mirror cells $(a, b), (b, a)$: equal under S;

under QS, $\pi_{ab}/\pi_{ba} = e^{\tau_b - \tau_a}$

Figure 10.2. The $I(I-1)/2$ mirror pairs of a square table ($I = 4$).

10.4.1 Symmetry as logit and loglinear models

When all $\pi_{ab} > 0$, symmetry is the logit model $\log(\pi_{ab}/\pi_{ba}) = 0$ for $a < b$, and, for expected frequencies $\mu_{ab} = n\pi_{ab}$, the loglinear model

$$\log \mu_{ab} = \lambda + \lambda_a + \lambda_b + \lambda_{ab}, \qquad \lambda_{ab} = \lambda_{ba}, \tag{10.7}$$

in which both classifications share the main effects; equivalently $\log \mu_{ab} = \lambda_{ab}$ with $\lambda_{ab} = \lambda_{ba}$.

Theorem 10.16 (Fitting and testing symmetry)

For Poisson or multinomial counts:
(a) the ML fit of symmetry is $\hat{\mu}_{ab} = (n_{ab} + n_{ba})/2$ (so $\hat{\mu}_{aa} = n_{aa}$);
(b) the residual df is $I(I-1)/2$;
(c) the Pearson statistic is Bowker's statistic

$$X^2 = \sum_{a<b} \frac{(n_{ab} - n_{ba})^2}{n_{ab} + n_{ba}},$$

which for $I = 2$ is McNemar's z_0^2;
(d) the standardized Pearson residuals are $r_{ab} = (n_{ab} - n_{ba})/(n_{ab} + n_{ba})^{1/2} = -r_{ba}$, and $\sum_{a<b} r_{ab}^2 = X^2$.

Proof.

Step 1. (a) Under $\log \mu_{ab} = \lambda_{ab}$ with $\lambda_{ab} = \lambda_{ba}$, the Poisson log-likelihood $\sum_{a,b}(n_{ab} \log \mu_{ab} - \mu_{ab})$ separates into terms $(n_{ab} + n_{ba}) \log \mu - 2\mu$ for each pair $a < b$ (with common value $\mu = \mu_{ab} = \mu_{ba}$) and $n_{aa} \log \mu_{aa} - \mu_{aa}$ for each diagonal cell. Maximizing each gives $\hat{\mu} = (n_{ab} + n_{ba})/2$ and $\hat{\mu}_{aa} = n_{aa}$. The multinomial fit is the same (the fitted total equals n).

Step 2. (b) There are I^2 cells and $I + \binom{I}{2}$ free parameters $\{\lambda_{ab}, a \le b\}$, so df $= I^2 - I - I(I-1)/2 = I(I-1)/2$. Equivalently, symmetry fixes the parameter of each of the $\binom{I}{2}$ binomials $n_{ab} \mid n_{ab} + n_{ba}$ at $\frac{1}{2}$.

Step 3. (c) Diagonal cells contribute 0. With $m = (n_{ab} + n_{ba})/2$, $(n_{ab} - m)^2 = (n_{ba} - m)^2 = (n_{ab} - n_{ba})^2/4$, so the pair contributes $2 \cdot \frac{(n_{ab}-n_{ba})^2}{4m} = \frac{(n_{ab}-n_{ba})^2}{n_{ab}+n_{ba}}$.

Step 4. (d) The Pearson residual of cell (a, b) is $(n_{ab} - m)/\sqrt{m}$. Each pair is fitted by one parameter with design column $(1, 1)'$ and equal weights, so the hat matrix for the pair is $\begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}$ and the leverage is $h = \frac{1}{2}$. Dividing by $\sqrt{1-h}$,

$$r_{ab} = \frac{(n_{ab} - n_{ba})/2}{\sqrt{m}\sqrt{1/2}} = \frac{n_{ab} - n_{ba}}{(n_{ab} + n_{ba})^{1/2}},$$

and summing r_{ab}^2 over $a < b$ reproduces (c). ■

Symmetry rarely fits (exceptions: intra-observer agreement); with substantial marginal heterogeneity it fits poorly.

10.4.2 Quasi-symmetry

Definition (Quasi-symmetry)

The **quasi-symmetry** (QS) model lets the main effects in (10.7) differ:

$$\log \mu_{ab} = \lambda + \lambda_a^X + \lambda_b^Y + \lambda_{ab}, \qquad \lambda_{ab} = \lambda_{ba}. \tag{10.8}$$

Symmetry is the case $\lambda_a^X = \lambda_a^Y$ for all a; independence is the case $\lambda_{ab} = 0$.

Since $\lambda_a^X + \lambda_b^Y = (\lambda_a^X + \lambda_b^X) + (\lambda_b^Y - \lambda_b^X)$ and the first bracket is symmetric, QS can be rewritten as

$$\log \mu_{ab} = \lambda'_{ab} + \tau_b, \qquad \lambda'_{ab} = \lambda'_{ba}, \quad \tau_b = \lambda_b^Y - \lambda_b^X, \quad \tau_I = 0. \tag{10.9}$$

Proposition 10.17 (Likelihood equations for quasi-symmetry)

The ML fit of QS satisfies

$$\hat\mu_{a+} = n_{a+}, \qquad \hat\mu_{+b} = n_{+b}, \qquad \hat\mu_{ab} + \hat\mu_{ba} = n_{ab} + n_{ba} \quad (a \le b);$$

one of the first two sets is implied by the other two. In particular $\hat\mu_{aa} = n_{aa}$, and the residual df is $(I-1)(I-2)/2$.

Proof.

Step 1. For a Poisson loglinear model the likelihood equations equate sufficient statistics to their expectations. In (10.8) the sufficient statistics are the row totals (λ^X), column totals (λ^Y) and the symmetric sums $n_{ab} + n_{ba}$, $a \le b$ (λ_{ab}; for $a = b$ the statistic is n_{aa}). Taking $a = b$ in the third set gives $\hat\mu_{aa} = n_{aa}$.

Step 2. Redundancy: given the third set and the row equations,

$$\hat\mu_{+a} = \sum_b \hat\mu_{ba} = \sum_b (n_{ab} + n_{ba} - \hat\mu_{ab}) = n_{a+} + n_{+a} - \hat\mu_{a+} = n_{+a}.$$

Step 3. df: in form (10.9) there are $I(I+1)/2$ parameters λ'_{ab} ($a \le b$) and $I-1$ parameters τ_b, so df $= I^2 - I(I+1)/2 - (I-1) = (I-1)(I-2)/2$. ∎

The equations have no closed-form solution; Newton–Raphson or IPF is used.

QS has the multiplicative form

$$\pi_{ab} = \alpha_a \beta_b \gamma_{ab}, \qquad \gamma_{ab} = \gamma_{ba}, \tag{10.10}$$

with positive parameters: the cellwise product of an independence table and a symmetric table. Symmetry is the case $\alpha_a = \beta_a$.

Theorem 10.18 (Odds-ratio characterization of quasi-symmetry)

Let all $\pi_{ab} > 0$.
(a) QS holds if and only if

$$\frac{\pi_{ab}\pi_{II}}{\pi_{aI}\pi_{Ib}} = \frac{\pi_{ba}\pi_{II}}{\pi_{bI}\pi_{Ia}} \qquad \text{for all } a, b. \tag{10.11}$$

(b) Under QS the local odds ratios satisfy $\theta_{ab} = \theta_{ba}$, where $\theta_{ab} = \pi_{ab}\pi_{a+1,b+1}/(\pi_{a,b+1}\pi_{a+1,b})$.

Proof. [Supplement]

Step 1. ($\Rightarrow$ in (a)) Substituting (10.10), the α's and β's cancel in each side, leaving $\gamma_{ab}\gamma_{II}/(\gamma_{aI}\gamma_{Ib})$ on the left and $\gamma_{ba}\gamma_{II}/(\gamma_{bI}\gamma_{Ia})$ on the right, which are equal by symmetry of γ.

Step 2. ($\Leftarrow$ in (a)) Put $\alpha_a = \pi_{aI}$, $\beta_b = \pi_{Ib}/\pi_{II}$ and $\gamma_{ab} = \pi_{ab}/(\alpha_a\beta_b) = \pi_{ab}\pi_{II}/(\pi_{aI}\pi_{Ib})$. Then $\pi_{ab} = \alpha_a\beta_b\gamma_{ab}$ by construction, and (10.11) says exactly $\gamma_{ab} = \gamma_{ba}$.

Step 3. (b) Substituting (10.10) into θ_{ab}, the α's and β's cancel, so $\theta_{ab} = \gamma_{ab}\gamma_{a+1,b+1}/(\gamma_{a,b+1}\gamma_{a+1,b})$; interchanging the roles of a and b and using $\gamma_{cd} = \gamma_{dc}$ gives the same value. ■

Goodman called QS the *symmetric association* model: odds ratios on one side of the diagonal mirror those on the other side.

The meaning of QS is clarified by subject-specific models. Generalize (10.6) to subject i:

$$\log \frac{P(Y_{it} = j)}{P(Y_{it} = I)} = \alpha_{ij} + \beta_j x_t, \qquad t = 1, 2, \quad j = 1, \ldots, I-1, \tag{10.12}$$

with $x_1 = 0$, $x_2 = 1$, $\alpha_{iI} = \beta_I = 0$, and responses independent given the subject.

Theorem 10.19 (Subject-specific logit models imply quasi-symmetry)

If (10.12) holds and subjects are sampled from any distribution F of $(\alpha_{i1}, \ldots, \alpha_{i,I-1})$, then the population-averaged table satisfies QS with $\tau_j = \lambda_j^Y - \lambda_j^X = \beta_j$ (constraint $\tau_I = 0$).

Proof. [Supplement]

Step 1. For a subject with parameter vector $\boldsymbol{\alpha}$, let $D_1(\boldsymbol{\alpha}) = \sum_k e^{\alpha_k}$ and $D_2(\boldsymbol{\alpha}) = \sum_k e^{\alpha_k + \beta_k}$. By conditional independence,

$$P(Y_1 = a, Y_2 = b \mid \boldsymbol{\alpha}) = \frac{e^{\alpha_a}}{D_1(\boldsymbol{\alpha})} \cdot \frac{e^{\alpha_b + \beta_b}}{D_2(\boldsymbol{\alpha})}.$$

Step 2. Averaging over F,

$$\pi_{ab} = e^{\beta_b}\,\gamma_{ab}, \qquad \gamma_{ab} = E_F\Big[\frac{e^{\alpha_a + \alpha_b}}{D_1(\boldsymbol{\alpha}) D_2(\boldsymbol{\alpha})}\Big],$$

and $\gamma_{ab} = \gamma_{ba}$ because $\alpha_a + \alpha_b$ is symmetric in (a, b).

Step 3. This is (10.10) with $\alpha_a \equiv 1$ and $\beta_b = e^{\beta_b}$, i.e. form (10.9) with $\tau_b = \beta_b$. ■

Conaway (1989) showed further that the conditional ML estimates of $\{\beta_j\}$ (conditioning out the α_{ij}) equal $\hat\lambda_j^Y - \hat\lambda_j^X$ from the ordinary ML fit of QS. Thus the main-effect differences of QS are subject-specific effects. QS also contains quasi-independence (Section 10.4.3) and the Bradley–Terry model (Section 10.6.3).

10.4.3 Quasi-independence

Square tables usually have more mass on the diagonal than independence predicts, but off the diagonal the structure may be simple.

Definition (Quasi-independence)

A square table satisfies **quasi-independence** (QI) if the classifications are independent given that the row and column outcomes differ:

$$\log \mu_{ab} = \lambda + \lambda_a^X + \lambda_b^Y + \delta_a\, I(a = b). \tag{10.13}$$

Each diagonal cell gets its own parameter, so the likelihood equations are $\hat{\mu}_{a+} = n_{a+}$, $\hat{\mu}_{+a} = n_{+a}$, $\hat{\mu}_{aa} = n_{aa}$: the diagonal is fitted perfectly and independence holds for the remaining cells. All odds ratios formed from four off-diagonal cells in a rectangle equal 1. The model has I more parameters than independence, so $\text{df} = (I-1)^2 - I$, and it applies for $I \geq 3$.

Proposition 10.20 (Quasi-independence within quasi-symmetry)

QI is the special case of QS in which the γ_{ab}, $a \neq b$, in (10.10) are all equal. For $I = 3$ the two models are equivalent.

Proof. [Supplement]

Step 1. Under QI, $\pi_{ab} = \alpha_a \beta_b$ for $a \neq b$ and $\pi_{aa} = \alpha_a \beta_a e^{\delta_a}$; this is (10.10) with $\gamma_{ab} = 1$ off the diagonal and $\gamma_{aa} = e^{\delta_a}$. Conversely, if the off-diagonal γ_{ab} equal a constant c, absorb c into α and the diagonal γ_{aa} into δ_a.

Step 2. Let $I = 3$ and $\pi_{ab} = \alpha_a \beta_b \gamma_{ab}$ with γ symmetric. We look for positive c_1, c_2, c_3 with $c_a c_b = \gamma_{ab}$ for $a \neq b$:

$$c_1 = \left(\frac{\gamma_{12}\gamma_{13}}{\gamma_{23}}\right)^{1/2}, \quad c_2 = \left(\frac{\gamma_{12}\gamma_{23}}{\gamma_{13}}\right)^{1/2}, \quad c_3 = \left(\frac{\gamma_{13}\gamma_{23}}{\gamma_{12}}\right)^{1/2}$$

satisfy $c_1 c_2 = \gamma_{12}$, $c_1 c_3 = \gamma_{13}$, $c_2 c_3 = \gamma_{23}$. Then $\pi_{ab} = (\alpha_a c_a)(\beta_b c_b)$ for $a \neq b$, which is QI with row and column parameters $\alpha_a c_a$, $\beta_b c_b$ and free diagonal. Hence QS $\subseteq$ QI when $I = 3$; with Step 1 the models coincide (both have df = 1). ■

10.4.4 Migration revisited

Example 10.21 (Migration: loglinear models)

For Example 10.15, independence fits terribly ($G^2 = 125{,}923$). Symmetry is also poor ($G^2 = 243.6$, df = 6): e.g. 124 people moved from NE to W but only 63 the reverse way. QI ($G^2 = 69.5$, df = 5) is much better, mainly because it fits the diagonal exactly, but it predicts too few NE→S and too many W→S moves. QS fits well: $G^2 = 3.0$, df = 3. Fitted values (diagonal fitted exactly by both):

	Quasi-independence				Quasi-symmetry			
1980	NE	MW	S	W	NE	MW	S	W
NE	–	126.6	312.9	150.5	–	95.8	370.4	123.8
MW	117.4	–	531.1	255.5	91.2	–	501.7	311.1
S	133.2	243.8	–	290.0	167.6	238.3	–	261.1
W	71.4	130.6	323.0	–	63.2	166.9	294.9	–

The subject-specific effects are $\hat{\tau}_1 = -0.672$, $\hat{\tau}_2 = -0.623$, $\hat{\tau}_3 = 0.122$ (W baseline). For a given person, the estimated odds of living in the South rather than the West in 1985 were $e^{0.122} = 1.13$ times the 1980 odds. Similar analyses apply to occupational mobility tables (parent's vs child's occupation).

10.4.5 Marginal homogeneity and quasi-symmetry

Marginal homogeneity is not a loglinear model, but QS gives a convenient route to testing it.

Theorem 10.22 (Caussinus)

For a table with all $\pi_{ab} > 0$,

$$\text{quasi-symmetry} + \text{marginal homogeneity} \iff \text{symmetry}.$$

Proof.

Step 1. ($\Leftarrow$) Symmetry is QS with $\alpha_a = \beta_a$, and implies marginal homogeneity.

Step 2. ($\Rightarrow$) Write QS as $\pi_{ab} = \alpha_a \beta_b \gamma_{ab}$ and set $\kappa_a = \alpha_a/\beta_a$, $\gamma^*_{ab} = \beta_a \beta_b \gamma_{ab}$, so that

$$\pi_{ab} = \kappa_a \gamma^*_{ab}, \qquad \gamma^*_{ab} = \gamma^*_{ba} > 0.$$

Step 3. Marginal homogeneity $\pi_{j+} = \pi_{+j}$ reads $\kappa_j \sum_b \gamma^*_{jb} = \sum_a \kappa_a \gamma^*_{aj}$, i.e.

$$\kappa_j = \sum_a \kappa_a \, w_{aj}, \qquad w_{aj} = \frac{\gamma^*_{aj}}{\sum_b \gamma^*_{bj}} > 0, \quad \sum_a w_{aj} = 1,$$

using $\gamma^*_{jb} = \gamma^*_{bj}$ in the denominator. So each κ_j is a weighted average, with strictly positive weights, of all of $\kappa_1, \ldots, \kappa_I$.

Step 4. Let j be an index with $\kappa_j = \min_a \kappa_a$. If some $\kappa_a > \kappa_j$, then the positive-weight average $\sum_a \kappa_a w_{aj}$ would exceed κ_j, a contradiction. Hence all κ_a equal a common κ.

Step 5. Then $\pi_{ab} = \kappa \gamma^*_{ab} = \kappa \gamma^*_{ba} = \pi_{ba}$: symmetry holds. ■

Corollary 10.23 (Conditional test of marginal homogeneity)

Given that QS holds, marginal homogeneity is equivalent to symmetry, i.e. to $\lambda^X_a = \lambda^Y_a$ (all $\tau_a = 0$). It is tested by

$$G^2(S \mid QS) = G^2(S) - G^2(QS), \qquad \text{df} = \tfrac{I(I-1)}{2} - \tfrac{(I-1)(I-2)}{2} = I - 1.$$

Proof. The equivalence is Theorem 10.22. In form (10.9), symmetry is the nested submodel $\tau_1 = \cdots = \tau_{I-1} = 0$, so the difference of deviances is the likelihood-ratio statistic, asymptotically χ^2 with df equal to the difference of the residual df of Theorem 10.16(b) and Proposition 10.17. ■

For the migration data, $G^2(S \mid QS) = 243.6 - 3.0 = 240.6$ (df $= 3$), close to the marginal-model statistics $G^2 = 240.8$ and $W = 236.5$ of Example 10.15.

10.4.6 Ordinal quasi-symmetry model

With ordered categories and scores $u_1 \le \cdots \le u_I$ for both classifications, the **ordinal quasi-symmetry** model is

$$\log \mu_{ab} = \lambda + \lambda_a + \lambda_b + \beta u_b + \lambda_{ab}, \qquad \lambda_{ab} = \lambda_{ba}, \tag{10.14}$$

the case of QS with linear trend $\tau_b = \lambda^Y_b - \lambda^X_b = \beta u_b$. Symmetry is $\beta = 0$. Dividing μ_{ab} by μ_{ba}, the symmetric terms cancel:

$$\log \frac{\pi_{ab}}{\pi_{ba}} = \beta(u_b - u_a), \qquad a < b. \tag{10.15}$$

This is a linear logit model with no intercept for the binomial $n_{ab} \mid (n_{ab} + n_{ba})$, with predictor $x = u_b - u_a$; it is fitted that way with logit software. The larger $|\beta|$, the larger the difference between π_{ab} and π_{ba}, hence between the margins.

Proposition 10.24 (Ordinal quasi-symmetry and marginal homogeneity)

Let $u_1 < \cdots < u_I$ and all $\pi_{ab} > 0$.
(a) The likelihood equations are $\sum_a u_a \hat{\mu}_{a+} = \sum_a u_a n_{a+}$, $\sum_b u_b \hat{\mu}_{+b} = \sum_b u_b n_{+b}$ and $\hat{\mu}_{ab} + \hat{\mu}_{ba} = n_{ab} + n_{ba}$ $(a \le b)$. The fitted margins need not equal the observed ones, but the fitted row and column means equal the observed means.
(b) Under (10.14), the column mean exceeds the row mean if $\beta > 0$ and is smaller if $\beta < 0$; hence ordinal QS + marginal homogeneity = symmetry, and $H_0 : \beta = 0$ tests marginal homogeneity (df = 1).

Proof. [Supplement]

Step 1. (a) The sufficient statistics of the loglinear model $\log \mu_{ab} = \lambda'_{ab} + \beta u_b$ (symmetric λ') are the symmetric sums and $\sum_b u_b n_{+b}$, which gives the second and third sets. The first set follows from the third:

$$\sum_{a,b}(u_a + u_b)\hat{\mu}_{ab} = \sum_{a \le b} c_{ab}(u_a + u_b)(\hat{\mu}_{ab} + \hat{\mu}_{ba}) = \sum_{a,b}(u_a + u_b) n_{ab},$$

where $c_{ab} = 1$ for $a < b$ and $c_{aa} = \frac{1}{2}$; subtracting the second set leaves $\sum_a u_a \hat{\mu}_{a+} = \sum_a u_a n_{a+}$. Dividing by n, the fitted row (column) mean equals the observed row (column) mean.

Step 2. (b) The difference between the column and row means is

$$\sum_{a,b}(u_b - u_a)\pi_{ab} = \sum_{a<b}(u_b - u_a)(\pi_{ab} - \pi_{ba}) = \sum_{a<b}(u_b - u_a)\pi_{ba}(e^{\beta(u_b - u_a)} - 1),$$

using (10.15). Every term has the sign of β (and all are nonzero when $\beta \neq 0$). Marginal homogeneity forces equal means, hence $\beta = 0$, hence symmetry; the converse is immediate. ■

The likelihood-ratio test compares the deviances of symmetry and ordinal QS. Like the ordinal marginal model, it concentrates the marginal comparison on a single df.

10.4.7 Premarital and extramarital sex revisited

Example 10.25 (Premarital and extramarital sex: symmetry models)

For Example 10.13, symmetry is hopeless ($G^2 = 402.2$, df = 6), while QS fits well ($G^2 = 1.4$, df = 3), and so does ordinal QS with scores $\{1, 2, 3, 4\}$ ($G^2 = 2.1$, df = 5), with $\hat{\beta} = -2.86$. By (10.15), the probability that the premarital response is x categories more permissive than the extramarital one is $e^{2.86x}$ times the reverse probability. For instance, "premarital almost always wrong, extramarital always wrong" is $e^{2.86} = 17.4$ times as likely as the reverse combination.

10.4.8 Other ordinal models for square tables

When symmetry fails for ordered classifications, often $\pi_{ab} > \pi_{ba}$ for all $a < b$ (or the reverse). The **conditional symmetry** model (McCullagh 1978)

$$\log(\pi_{ab}/\pi_{ba}) = \tau \qquad \text{for all } a < b \tag{10.16}$$

captures this; it implies that, given $Y_1 \neq Y_2$, the probabilities above the diagonal are a mirror image (up to the factor e^{τ}) of those below. Symmetry is $\tau = 0$.

The **quasi-uniform association** model generalizes quasi-independence:

$$\log \mu_{ab} = \lambda + \lambda_a^X + \lambda_b^Y + \beta u_a u_b + \delta_a I(a = b), \tag{10.17}$$

linear-by-linear association off the diagonal. It is a special case of QS (the term $\beta u_a u_b$ is symmetric), and QI is $\beta = 0$; with equal-interval scores the local odds ratios off the diagonal all equal e^{β}.

For Table 10.13's data, conditional symmetry gives $\hat{\tau} = -4.130$ (SE $= 0.451$): a more negative opinion on extramarital than on premarital sex is $e^{4.13} = 62.2$ times as likely as the reverse. Quasi-uniform association gives $\hat{\beta} = 0.632$ (SE $= 0.106$), i.e. an off-diagonal local odds ratio of $e^{0.632} = 1.88$.

10.5 Measuring agreement between observers

Motivation. When two raters classify the same subjects, the square table describes their agreement. Agreement is a different property from association, so we need models (quasi-independence, quasi-symmetry) and summary indices (kappa) aimed specifically at the diagonal.

Example 10.26 (Pathologists' diagnoses)

Two pathologists A and B classified 118 slides on carcinoma of the uterine cervix: 1 = negative, 2 = atypical squamous hyperplasia, 3 = carcinoma in situ, 4 = squamous or invasive carcinoma. In parentheses: standardized Pearson residuals for independence.

	Pathologist B				
Pathologist A	1	2	3	4	Total
1	22 (8.5)	2 (−0.5)	2 (−5.9)	0 (−1.8)	26
2	5 (−0.5)	7 (3.2)	14 (−0.5)	0 (−1.8)	26
3	0 (−4.1)	2 (−1.2)	36 (5.5)	0 (−2.3)	38
4	0 (−3.3)	1 (−1.3)	17 (0.3)	10 (5.9)	28
Total	27	12	69	10	118

Independence fits poorly ($G^2 = 118.0$, df $= 9$). Large positive residuals on the diagonal show agreement beyond chance in every category, especially the first; off-diagonal residuals are mostly negative, less so for adjacent categories. The most common disagreements: B chose 3 while A chose 2 or 4.

10.5.1 Agreement: departures from independence

Let π_{ab} be the probability that A classifies a subject in a and B in b. Then $\sum_a \pi_{aa}$ is the probability of agreement, and perfect agreement is $\sum_a \pi_{aa} = 1$. Strong agreement requires strong association, but not conversely: if A always rates one category higher than B, association is strong and agreement poor. Independence is the baseline of "chance" agreement; its standardized residuals reveal patterns of agreement and disagreement (they grow with n, so their size is not a measure of agreement).

10.5.2 Using quasi-independence to analyze agreement

QI (10.13) adds agreement parameters δ_a to independence. For two subjects, suppose each observer puts one in category a and the other in b. The odds that the observers agree rather than disagree on which is which are

$$\tau_{ab} = \frac{\pi_{aa}\pi_{bb}}{\pi_{ab}\pi_{ba}} = \frac{\mu_{aa}\mu_{bb}}{\mu_{ab}\mu_{ba}}. \tag{10.18}$$

Proposition 10.27 (Agreement odds under QI and QS)

Under QI, $\tau_{ab} = \exp(\delta_a + \delta_b)$. Under QS, $\tau_{ab} = \exp(\lambda_{aa} + \lambda_{bb} - \lambda_{ab} - \lambda_{ba})$.

Proof. [Supplement] Under QI, $\log \mu_{aa} + \log \mu_{bb} - \log \mu_{ab} - \log \mu_{ba}$ equals

$$(\lambda_a^X + \lambda_a^Y + \delta_a) + (\lambda_b^X + \lambda_b^Y + \delta_b) - (\lambda_a^X + \lambda_b^Y) - (\lambda_b^X + \lambda_a^Y) = \delta_a + \delta_b.$$

Under QS the same computation with (10.8) cancels all main effects and leaves $\lambda_{aa} + \lambda_{bb} - \lambda_{ab} - \lambda_{ba}$. ■

Larger δ_a mean stronger agreement. For Example 10.26, QI has $G^2 = 13.2$ (df = 5), much better than independence but with some lack of fit; $\hat{\delta}_2 = 0.6$, $\hat{\delta}_3 = 1.9$ and $\hat{\tau}_{23} = e^{2.5} = 12.3$.

10.5.3 Quasi-symmetry and agreement modeling

Given disagreement, some association often remains, which QI forbids. QS permits it, and for Example 10.26 it fits well: $G^2 = 1.0$, df = 2. When $n_{ab} + n_{ba} = 0$ (here categories 1 and 4), the QS likelihood equation $\hat{\mu}_{ab} + \hat{\mu}_{ba} = n_{ab} + n_{ba}$ forces both fitted values to 0; those cells should be removed from the fit to get the correct df. Under QS, $\hat{\tau}_{23} = 10.7$.

	Quasi-independence				Quasi-symmetry			
A	1	2	3	4	1	2	3	4
1	22	0.7	3.3	0.0	22	2.4	1.6	0.0
2	2.4	7	16.6	0.0	4.6	7	14.4	0.0
3	0.8	1.2	36	0.0	0.4	1.6	36	0.0
4	1.9	3.0	13.1	10	0.0	1.0	17.0	10

Fitted values for Example 10.26.

QS also informs about the margins: symmetry fits poorly ($G^2 = 39.2$, df = 5), and $G^2(S \mid QS) = 39.2 - 1.0 = 38.2$ (df = 3) shows strong marginal heterogeneity, one reason agreement is not stronger. Ordinal versions take the category order into account: given disagreement, high (low) ratings by one observer still tend to go with relatively high (low) ratings by the other.

10.5.4 Kappa measure of agreement

Definition (Cohen's kappa)

With $P_o = \sum_a \pi_{aa}$ and $P_e = \sum_a \pi_{a+}\pi_{+a}$ (agreement expected under independence),

$$\kappa = \frac{\sum_a \pi_{aa} - \sum_a \pi_{a+}\pi_{+a}}{1 - \sum_a \pi_{a+}\pi_{+a}} = \frac{P_o - P_e}{1 - P_e}.$$

The denominator is the numerator with P_o replaced by its maximum 1. So $\kappa = 0$ when agreement equals chance agreement, $\kappa = 1$ under perfect agreement, and $\kappa < 0$ (rare) when agreement is worse than chance. For given margins, stronger agreement means larger κ.

Theorem 10.28 (Asymptotic variance of sample kappa)

Under multinomial sampling, $\sqrt{n}(\hat{\kappa} - \kappa) \xrightarrow{d} N(0, \sigma_\kappa^2)$, where

$$\sigma_\kappa^2 = \frac{P_o(1-P_o)}{(1-P_e)^2} + \frac{2(1-P_o)[2P_oP_e - \sum_a \pi_{aa}(\pi_{a+} + \pi_{+a})]}{(1-P_e)^3} + \frac{(1-P_o)^2[\sum_a \sum_b \pi_{ab}(\pi_{b+} + \pi_{+a})^2 - 4P_e^2]}{(1-P_e)^4}.$$

The estimated variance of $\hat{\kappa}$ (Fleiss, Cohen and Everitt 1969) substitutes sample proportions and divides by n.

Proof. [Supplement]

Step 1. Delta method for a multinomial. If $\sqrt{n}(\mathbf{p}-\boldsymbol{\pi}) \xrightarrow{d} N(\mathbf{0}, \operatorname{diag}(\boldsymbol{\pi})-\boldsymbol{\pi}\boldsymbol{\pi}')$ and g is differentiable with gradient (g_{ab}), then $\sqrt{n}(g(\mathbf{p})-g(\boldsymbol{\pi})) \xrightarrow{d} N(0,\sigma^2)$ with

$$\sigma^2 = \sum_{a,b} \pi_{ab} g_{ab}^2 - \Big(\sum_{a,b} \pi_{ab} g_{ab}\Big)^2.$$

Step 2. Gradient. Write $\kappa = 1-(1-P_o)/(1-P_e)$. Since π_{ab} enters π_{a+} and π_{+b}, $\partial P_e/\partial \pi_{ab} = \pi_{+a}+\pi_{b+}$, and $\partial P_o/\partial \pi_{ab} = I(a=b)$. Hence

$$g_{ab} = \frac{I(a=b)(1-P_e) - (1-P_o)(\pi_{+a}+\pi_{b+})}{(1-P_e)^2}.$$

Step 3. First moment. Since $\sum_{a,b}\pi_{ab}(\pi_{+a}+\pi_{b+}) = \sum_a \pi_{a+}\pi_{+a} + \sum_b \pi_{+b}\pi_{b+} = 2P_e$,

$$\sum_{a,b} \pi_{ab} g_{ab} = \frac{P_o(1-P_e) - 2P_e(1-P_o)}{(1-P_e)^2}.$$

Step 4. Second moment. Let $s = \sum_a \pi_{aa}(\pi_{a+}+\pi_{+a})$ and $t = \sum_{a,b}\pi_{ab}(\pi_{+a}+\pi_{b+})^2$. Expanding the square,

$$(1-P_e)^4 \sum_{a,b} \pi_{ab} g_{ab}^2 = P_o(1-P_e)^2 - 2(1-P_e)(1-P_o)s + (1-P_o)^2 t.$$

Step 5. Subtract the square of Step 3, $(1-P_e)^{-4}[P_o^2(1-P_e)^2 - 4P_oP_e(1-P_e)(1-P_o) + 4P_e^2(1-P_o)^2]$:

$$(1-P_e)^4\sigma_\kappa^2 = P_o(1-P_o)(1-P_e)^2 + 2(1-P_e)(1-P_o)(2P_oP_e - s) + (1-P_o)^2(t-4P_e^2).$$

Dividing by $(1-P_e)^4$ gives the stated formula. ∎

It is rarely plausible that agreement is only at chance level, so instead of testing $\kappa = 0$ one estimates κ by an interval. For Example 10.26, $P_o = 0.636$, $P_e = 0.281$, $\hat{\kappa} = (0.636-0.281)/(1-0.281) = 0.493$ and SE $= 0.057$: the excess of observed over chance agreement is about half its maximum possible value, and κ lies roughly between 0.4 and 0.6.

10.5.5 Weighted kappa: quantifying disagreement

Definition (Weighted kappa)

With weights $0 \le w_{ab} \le 1$, $w_{aa} = 1$, $w_{ab} = w_{ba}$,

$$\kappa_w = \frac{\sum_a \sum_b w_{ab}\pi_{ab} - \sum_a \sum_b w_{ab}\pi_{a+}\pi_{+b}}{1 - \sum_a \sum_b w_{ab}\pi_{a+}\pi_{+b}}.$$

Common choices are $w_{ab} = 1-|a-b|/(I-1)$ and (Fleiss–Cohen) $w_{ab} = 1-(a-b)^2/(I-1)^2$.

Weights let disagreements between distant categories count as more serious. Both kappa and weighted kappa depend strongly on the marginal distributions: the same rating process can give very different values for populations with different case mixes. A single index loses much information; models describing the agreement and disagreement structure are more informative.

10.5.6 Extensions to multiple observers

With R observers, ordinary loglinear models describe pairwise agreement *conditional* on the other ratings, which is usually not what is wanted. It is more natural to model the $\binom{R}{2}$ two-way marginal tables simultaneously (Becker and Agresti 1992), to use generalized kappas summarizing pairwise or multiple agreement, or to use latent-class mixture models with classes of subjects on whom observers agree or disagree (Chapter 13).

10.6 Bradley–Terry model for paired preferences

Motivation. When items (teams, wines, products) are compared two at a time, a single ranking can be built from the pairwise outcomes. The Bradley–Terry model is a logit model for these comparisons and is, in fact, quasi-symmetry in disguise.

10.6.1 Bradley–Terry model

Let Π_{ab} be the probability that a is preferred to b, with $\Pi_{ab} + \Pi_{ba} = 1$ (no ties).

Definition (Bradley–Terry model)

$$\log \frac{\Pi_{ab}}{\Pi_{ba}} = \beta_a - \beta_b, \qquad \text{equivalently} \qquad \Pi_{ab} = \frac{e^{\beta_a}}{e^{\beta_a} + e^{\beta_b}}. \tag{10.19}$$

Identifiability needs a constraint such as $\beta_I = 0$ or $\sum_a e^{\beta_a} = 1$.

So $\Pi_{ab} = \frac{1}{2}$ when $\beta_a = \beta_b$ and $\Pi_{ab} > \frac{1}{2}$ when $\beta_a > \beta_b$. The model describes $\binom{I}{2}$ probabilities by $I-1$ parameters, so residual df $= \binom{I}{2} - (I-1)$. Let N_{ab} be the number of comparisons of a and b, with a preferred n_{ab} times and b preferred $n_{ba} = N_{ab} - n_{ab}$ times; the results form a square table with empty diagonal. If comparisons are independent, $n_{ab} \sim \text{bin}(N_{ab}, \Pi_{ab})$ independently over pairs, and the model is an ordinary no-intercept logit model whose predictor row for pair (a, b) has $+1$ in position a and -1 in position b.

Proposition 10.29 (Likelihood equations for Bradley–Terry)

The ML estimates satisfy, for each item a,

$$\sum_{b \neq a} N_{ab} \hat{\Pi}_{ab} = \sum_{b \neq a} n_{ab} :$$

the fitted number of wins of each item equals its observed number of wins.

Proof. [Supplement] The log-likelihood is $L = \sum_{a<b} \left[n_{ab}(\beta_a - \beta_b) - N_{ab} \log(1 + e^{\beta_a - \beta_b}) \right]$. Item a appears with sign $+1$ in pairs (a, b), $b > a$, and with sign -1 in pairs (c, a), $c < a$. Differentiating,

$$\frac{\partial L}{\partial \beta_a} = \sum_{b>a} (n_{ab} - N_{ab}\Pi_{ab}) - \sum_{c<a} (n_{ca} - N_{ca}\Pi_{ca}) = \sum_{b \neq a} (n_{ab} - N_{ab}\Pi_{ab}),$$

since $n_{ca} - N_{ca}\Pi_{ca} = -(n_{ac} - N_{ac}\Pi_{ac})$. Setting this to zero gives the result. ∎

10.6.2 Home team advantage in baseball

Example 10.30 (Baseball, 1987 American League East)

Wins of the row team over the column team (Bradley–Terry fitted values in parentheses):

Winner	Milw.	Detr.	Toro.	N.Y.	Bost.	Clev.	Balt.
Milwaukee	–	7 (7.0)	9 (7.4)	7 (7.6)	7 (8.0)	9 (9.2)	11 (10.8)
Detroit	6 (6.0)	–	7 (7.0)	5 (7.1)	11 (7.6)	9 (8.8)	9 (10.5)
Toronto	4 (5.6)	6 (6.0)	–	7 (6.7)	7 (7.1)	8 (8.4)	12 (10.2)
New York	6 (5.4)	8 (5.9)	6 (6.3)	–	6 (7.0)	7 (8.3)	10 (10.1)
Boston	6 (5.0)	2 (5.4)	6 (5.9)	7 (6.0)	–	7 (7.9)	12 (9.8)
Cleveland	4 (3.8)	4 (4.2)	5 (4.6)	6 (4.7)	6 (5.1)	–	6 (8.6)
Baltimore	2 (2.2)	4 (2.5)	1 (2.8)	3 (2.9)	1 (3.2)	7 (4.4)	–

Fitting (10.19) to the $\binom{7}{2} = 21$ binomials gives $G^2 = 15.7$ (df $= 15$). Estimates ($\hat\beta_7 = 0$):

	Milw.	Detr.	Toro.	N.Y.	Bost.	Clev.	Balt.
Winning %	64.1	60.2	56.4	55.1	51.3	39.7	23.1
$\hat\beta_a$	1.58	1.44	1.29	1.25	1.11	0.68	0.00
$e^{\hat\beta_a}$, $\sum = 1$	0.218	0.189	0.164	0.158	0.136	0.089	0.045

For Boston vs New York, $\hat\Pi_{54} = e^{1.11}/(e^{1.11} + e^{1.25}) = 0.46$. Each $\hat\beta_a$ and $\hat\beta_a - \hat\beta_b$ has SE about 0.3, so the top five teams are not clearly distinguished.

Home advantage. Let Π^*_{ab} be the probability that a beats b when a is at home, and fit

$$\log \frac{\Pi^*_{ab}}{1 - \Pi^*_{ab}} = \alpha + (\beta_a - \beta_b) \tag{10.20}$$

to the 42 home/away binomials (e.g. Boston at home beat New York 4–2; New York at home beat Boston 4–3). This gives $G^2 = 38.6$ (df $= 35$), team estimates similar to before ($e^{\hat\beta_a}$: $0.220, 0.190, 0.164, 0.157, 0.137, 0.088, 0.044$) and $\hat\alpha = 0.302$: the home team of two evenly matched teams wins with probability $e^{0.302}/(1+e^{0.302}) = 0.575$. Boston beats New York with estimated probability 0.54 at Boston and 0.39 at New York.

Model (10.20) is useful whenever there is an order effect, e.g. the product tasted first in a taste test.

10.6.3 Bradley–Terry model and quasi-symmetry

Theorem 10.31 (Fienberg–Larntz)

Let the square table of comparisons have cell (a, b) = "a preferred to b". If the table satisfies QS (10.8), then the conditional probability of cell (a, b) given that the observation is in (a, b) or (b, a) follows the Bradley–Terry model with $\beta_a = \lambda^X_a - \lambda^Y_a$. Conversely, the Bradley–Terry model for $\{\Pi_{ab}\}$ is the logit form of QS, and the ML estimates $\hat\lambda^X_a - \hat\lambda^Y_a$ from QS give $\hat\beta_a$.

Proof.

Step 1. Under QS,

$$\begin{aligned}\log \frac{\mu_{ab}}{\mu_{ba}} &= (\lambda + \lambda^X_a + \lambda^Y_b + \lambda_{ab}) - (\lambda + \lambda^X_b + \lambda^Y_a + \lambda_{ba}) \\ &= (\lambda^X_a - \lambda^Y_a) - (\lambda^X_b - \lambda^Y_b) = \beta_a - \beta_b,\end{aligned}$$

since $\lambda_{ab} = \lambda_{ba}$; and $\mu_{ab}/(\mu_{ab} + \mu_{ba})$ is the conditional probability of (a, b) given the pair.

Step 2. Conversely, the QS likelihood, conditional on the pair totals $n_{ab} + n_{ba} = N_{ab}$ (which are the sufficient statistics for the symmetric terms and are fitted exactly by Proposition 10.17), is the product of binomials with logits $\beta_a - \beta_b$, i.e. the Bradley–Terry likelihood. Since the symmetric parameters are fitted by $\hat{\mu}_{ab} + \hat{\mu}_{ba} = N_{ab}$ and the remaining factor depends only on $\{\beta_a\}$, the two fits give the same $\hat{\beta}_a$. ■

10.6.4 Extensions to ties and ordinal evaluations

For an ordinal evaluation scale with I categories (e.g. much better, slightly better, same, slightly worse, much worse), let Y_{ab} be the response comparing a with b and use

$$\operatorname{logit} P(Y_{ab} \le j) = \theta_j + \beta_a - \beta_b.$$

Proposition 10.32 (Symmetry of the cutpoints)

Since comparing b with a reverses the scale, $Y_{ba} = I+1-Y_{ab}$, and then necessarily $\theta_j = -\theta_{I-j}$. For the scale (win, tie, lose), $\theta_1 = -\theta_2$.

Proof. $P(Y_{ab} \le j) = P(Y_{ba} \ge I+1-j) = 1 - P(Y_{ba} \le I-j)$, so $\operatorname{logit} P(Y_{ab} \le j) = -\operatorname{logit} P(Y_{ba} \le I-j)$. Substituting the model, $\theta_j + \beta_a - \beta_b = -(\theta_{I-j} + \beta_b - \beta_a) = -\theta_{I-j} + \beta_a - \beta_b$, so $\theta_j = -\theta_{I-j}$. ■